

**Lab Goal :** This lab was designed to teach you more about recursion.

**Lab Description :** Write a program that takes in a string of unique characters and generates all possible permutations of the letters.

### Example showing the recursion:

```

p("CAT")
init = "C"
words = p("AT")
      = {"AT", "TA"}
Insert "C" at every spot
in "AT" and "TA" to form
"CAT", "ACT", "ATC",
"CTA", "TCA", and "TAC"
return answer

p("AT")
init = "A"
words = p("T")
      = {"T"}
Insert "A" at every spot
in "T" to form "AT"
and "TA"
return {"AT", "TA"}

p("T")
init = "T"
words = p("")
      = {""}
Insert "T" in front
of "" to form "T"
return {"T"}

p("")
return {""}

```

### Sample Data :

ABC  
Abc  
Boat  
it

### Sample Output :

**Permutations of CAT are:**

ATC CTA ACT TCA CAT TAC

**Permutations of abc are:**

bca acb abc cba bac cab

**Permutations of boat are:**

tbao baot btoa toab abot atob tabo toba aobt atbo boat btao otba aotb abto  
oatb obta bato bota obat taob tboa oabt

**Permutations of frown are:**

wforn nforw nfwro fnwor ofwnr nrwof onwfr wofnr wornf fnowr nwrfo ronfw  
ronwf fonrw nfrow wfnor rfon rwnof wfron norfw nofwr norwf worfn wonfr  
nwfor rofnw wnfro nfrwo rwnfo wrfno wfrno nrfwo rwfno wrnfo nworf wfnro  
wnrof wnfor rfnow nwofr rowfn frnow fowrn nowfr rfnwo rownf rnfo wofrn  
wnorf rwofn wronf fwnro rnfwo nrowf rwnof wrofn nrwfo nrofw ownfr fornw  
owrfn nofrw wofrn wonrf owfnr rfwon ownrf fwrno frwon onfrw rfown frnwo  
orfnw ornwf ornfw fworn ofnrw forwn fonwr rfwno rfonw fwrn fwnor rnowf  
rnwfo rnofw fownr fnrow wrnof wrfon nowrf ofrnw nrfow onrfw onfwr onrnf  
fwonr nfwor ofrwn ofnwr onwrf frown nfowr orwfn ofwrn wfonr orwnf fnowr  
frwno fronw nwfro

### algorithm help

Make a HashSet to store the permutations of the word

Handling error scenarios

If the word is nonexistent or the empty string:

If the word is the empty string, add it to your set

Return your set

Store the first character

Store the rest of the string

Call permutations() on the remainder and store the set it gives you

Loop through each permutation in the set

Loop through each spot of the current word from rem

Insert the initial letter at the current spot

Add this permutation to your set of permutations.

Return the set of permutations

**Test `null` to make sure you're writing robust programs.**

**Your program should give a good error message.**